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Exponential family distribution and sufficient statistic.

Asked 5 years, 11 months ago Modified 4 months ago Viewed 3k times



The exponential distribution family is defined by a PDF of the form:

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$$f_X(x; \theta) = c(\theta)g(x)\exp\left[\sum_{j=1}^l G_j(\theta)T_j(x)\right],$$



where $\theta \in \Theta$ and $c(\theta) > 0$, $Q_j(\theta)$ are arbitrary functions of θ , and $g(x) > 0$ and $t(x)$ are arbitrary functions of x .



This seems very complicated to an untrained eye and honestly, I don't think I understand it.

I researched the mighty internet and found a simplified form:

$$f(x) = \exp\left[\frac{\theta(x) - b(\theta)}{a(\Phi)} + c(x, \Phi)\right],$$

which seems much more user friendly for beginners.

However, there is an extension to the first exponential family PDF definition, such that, by applying the factorization theorem to the joint PDF $f_X(\mathbf{x}; \theta)$, one obtains the **sufficient statistic**:

$$\mathbf{T}(\mathbf{X}) = \left(\sum_{i=1}^n T_1(X_i), \dots, \sum_{i=1}^n T_l(x_i) \right),$$

which is a sufficient statistic for $Q_1(\theta), \dots, Q_l(\theta)$.

How can the sufficient statistic be obtained from the simplified version of the exponential family form?

How can variance and mean be calculated from the first definition of the exponential family form?

EXAMPLE:

Prove that Poisson distribution belongs to the exponential family.

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Exercise 10. Prove that the Poisson distribution belongs to the exponential family.

Poisson model: $X \sim Ps(\theta)$, $\theta \in \Theta = \mathbb{R}_+ \setminus 0$ The pdf is given by

$$\begin{aligned} f_X(x; \theta) &= e^{-\theta} \frac{\theta^x}{x!} \mathbb{1}_{\mathbb{N}}(x) \\ &= e^{-\theta} \frac{1}{x!} e^{x \ln(\theta)} \mathbb{1}_{\mathbb{N}}(x) \end{aligned}$$

where $\mathbb{1}_{\mathbb{N}}(x) = 1$ if $x \in \mathbb{N}$ and zero elsewhere.

We conclude that Poisson model is member of exponential family since

$$\begin{aligned} c(\theta) &= e^{-\theta} \\ g(x) &= \frac{1}{x!} \mathbb{1}_{\mathbb{N}}(x) \\ T(x) &= x \quad (\ell = 1) \\ Q(\theta) &= \ln(\theta) \end{aligned}$$



Question: How can $E[X]$ and $\text{Var}[X]$ be calculated here?

I was able to do this with the simplified form that I found online, as follows:

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

and by taking the log and then the exponential on both sides one gets:

$$f(x) = \exp[x \log(\lambda) - \lambda - \log(x!)]$$

Matching this expression to the simplified form of the exponential family, we get: $\theta = \log(\lambda)$

$$\lambda = \exp(\theta)$$

$$b(\theta) = e^{\theta}$$

$a(\Phi)$, always 1 for distributions with one parameter

$$c(x, \Phi) = -\log(x!)$$

Then one can easily get:

$$\begin{aligned} E[X] &= b'(\theta) = \lambda \\ \text{Var}[X] &= a(\Phi) b''(\theta) = \lambda \end{aligned}$$

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edited Apr 5 at 16:42



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59 1 1 10

asked Aug 29, 2018 at 10:33



user1607

481 6 24



Many typos in the question. – [Michael](#) Jul 27, 2019 at 23:20

1 Answer

Sorted by: [Highest score \(default\)](#)



First question

1

Inspecting the definition of the exponential family



$$f_x(x; \theta) = c(\theta)g(x)e^{\sum_{j=1}^l G_j(\theta)T_j(x)},$$



one can say the following:



1. T is a sufficient statistic. Condition on T , the conditional distribution is $g(x)$ (up to a normalization constant), which is independent of the parameter θ . This is the definition of sufficiency. In fact, for the exponential family it is independent of T .
2. The term $e^{\sum_{j=1}^l G_j(\theta)T_j(x)}$ determines the marginal distribution of T , via the choice of G_j 's.
3. $c(\theta)$ is a normalization constant so the density integrates to 1.

Second question

As G_j 's are arbitrary, subject to measurability requirements etc., there is no general formula for computing moments. For the Poisson distribution, the first moment is simply

$$e^{-\lambda} \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} = (e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}) \cdot \lambda = \lambda.$$

The second moment is similar.

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answered Jul 27, 2019 at 23:20



Michael

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